



KAMPALA INTERNATIONAL UNIVERSITY-WESTERN CAMPUS
FACULTY OF BIOMEDICAL SCIENCES
DEPARTMENT OF PHYSIOLOGY
END OF BIOMEDICALS EXAMS (EBE)
MEDICAL PHYSIOLOGY II (CVS, RESP, ENDO & REPRO) MPH 120
BMS 1.2

*Agreeing to me
by [signature] 11/11*

DATE: 22ND NOV. 2019

(ACADEMIC YEAR: 2018/2019)

TIME 150 MINS

READ THE INSTRUCTIONS CAREFULLY: This paper consists of 3 sections,

SECTION A: which you are to answer on the MCQ sheets provided at the front of the question paper.

Section B: are short answer questions, which you are to answer in the spaces, table or graph provided only.

Section C: is essay section and you are to answer ALL questions.

SECTION A: MULTIPLE CHOICE QUESTION (MCQ) CHOOSE THE MOST CORRECT SINGLE OPTION ONLY. (50 MARKS TOTAL)

CVS

1. The following cardiovascular changes take place during exercise except -----

- A. Production of Red Blood Cells due to hypoxia
- B. Reduced blood volume due to fluid loss
- ☒ C. Decreased blood flow to the skeletal muscles
- D. Increased cardiac output

A device which measures electrical activity of the heart is called-----

- A. Electrocardiogram
- ☒ B. Electrocardiograph
- C. Sphygmomanometer
- D. Stethoscope

The parasympathetic regulation of cardiac output includes the following except-----

- ☒ A. Release of norepinephrine
- B. Release of acetylcholine
- C. Increased K⁺ channel opening
- D. Decreased A-V node transmission

In pulmonary circulation, which blood vessel carries the oxygenated blood?

- A. Pulmonary artery
- B. Coronary artery
- C. Coronary vein
- ☒ D. Pulmonary vein

Depolarization of a contractile myocardial cell is associated with----

- ☒ A. Sodium influx
- B. Potassium influx
- C. Calcium influx
- D. Chloride efflux

6. QRS complex in a typical electrocardiogram denotes -----

- A. Atrial depolarization
- ☒ B. Atrial repolarization and ventricular depolarization
- C. Ventricular repolarization
- D. Ventricular depolarization

7. ----- is a long-term mechanism for the regulation of blood pressure

- ☒ A. The renal mechanism
- B. The baroreceptor mechanism
- C. The chemoreceptor mechanism
- D. The nervous mechanism

In event where there is a sphygmomanometer but no stethoscope, what method of blood pressure measurement is clinically most appropriate?

- A. Auscultatory method
- B. Oscillatory method
- ☒ C. Palpatory method
- D. Physiographic method

8. All these affects Frank-Starling mechanism except-----

- A. Stroke volume
- B. Venous return
- C. Ventricular contraction
- ☒ D. Blood volume

10. A myocardial cell exhibit the following properties except-----

- ☒ A. Rigidity
- B. Automaticity

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C. Contractility

D. Excitability

In the standard limb leads, Lead 1 represents a connection between-----

- A. Right arm and right foot
- ☒ B. Right arm and left arm

- C. Left arm and left foot
- D. Right arm and left foot

-----explains Kirchhoff's Law

- A. $V_1 + V_{11} = V_{111}$
- ☒ B. $V_i + V_{111} = V_{11}$

- C. $V_{11} + V_{111} = V_1$
- D. $V_1 + V_{11} = V_0$

The following health disorders can result in ST segment elevation except-----

- A. Hyperkalemia
- ☒ B. Hypokalemia

- C. Acute myocardial infarction
- D. Left ventricular hypertrophy

In systemic circulation, blood flow begins from-----

- ☒ A. Left atrium
- B. Right atrium

- C. Right ventricle
- D. Left ventricle

Pulmonary circulation begins from-----

- A. Right ventricle
- B. Left ventricle

- C. Left atrium
- ☒ D. Right atrium

Electrical activities of the heart begins from-----

- ☒ A. The sinoatrial node
- B. The atrioventricular node

- C. Purkinje fibres
- D. Bundle of HIS

Inflammation of the muscular layer of the heart is termed-----

- A. Myocardium infarction
- ☒ B. Myocarditis

- C. Pericarditis
- D. Epicarditis

What is the mean arterial pressure of a subject with a blood pressure of 120/90mmHg

- ☒ A. 100mmHg
- B. 93.3mmHg

- C. 103mmHg
- D. 93mmHg

One of these is not true of the SA node

- A. It contains specialized cells that do not contract
- ☒ B. It is located in the left atrium

- C. It is responsible for the generation of heart rhythm
- D. It is called the pacemaker

Veins differ from arteries in that:

- A. They have elastic walls
- B. They carry blood away from the heart

- C. They transport oxygenated blood
- ☒ D. They carry blood to the heart

Given that the systolic pressure is 135mmHg, and the Diastolic pressure is 85mmHg, what is the pulse pressure?

- A. 55mmHg
- ☒ B. 50mmHg

- C. 220mmHg
- D. 110mmHg

The following does not describe the arteries except-----

- A. They form a link between the arterioles and venules
- B. They form a site for gaseous and nutrient exchange

- C. They have a low pressure load
- ☒ D. Their walls are thick and elastic

Coronary blood flow is affected by the following factors except-----

- A. Nervous factors
- ☒ B. Metabolic factors

- C. Need for oxygen
- ☒ D. Endocrine factors

Baroreceptors are found in -----

- A. Carotid body
- ☒ B. Carotid Sinus

- C. Aortic body
- D. Femoral Sinus

Chemoreceptors can be found in the -----

- A. Carotid sinus
- ☒ B. Carotid body

- C. Aortic sinus
- D. Femoral body

RESPIRATION

All of the followings are factors that affect the compliance of the lungs EXCEPT:

- a. ~ Radius of the alveolus
- c. ~ Pulmonary Surfactant

- ☒ C. Solubility of the gases
- d. ~ Elastic fibers in the alveoli

Which of the following lung volumes cannot be recorded by spirometry?

c. inspiratory reserve

d. expiratory reserve

28. The vital capacity includes all the following volumes EXCEPT:

- a. tidal volume
- ☒ b. residual volume

- c. inspiratory reserve volume
- d. expiratory reserve volume

29. Anatomical dead space:

- a. is absent in healthy individuals
- b. is greater than physiological dead space

- ☒ c. is found in the upper airways
- d. includes the gas exchange areas

30. The normal value for the oxygen partial pressure in the humidified air at sea level is:

- a. 95 mmHg
- b. 104 mmHg

- ☒ c. 149 mmHg
- d. 159 mmHg

31. Which of the following factors DOES NOT affect the composition of the alveolar air?

- a. amount and quality of air delivered to and from the alveoli by ventilation
- b. diffusion rate of gases through the respiratory membrane
- ☒ c. affinity of the haemoglobin for the respiratory gases
- d. blood flow through pulmonary capillaries

32. The diffusing capacity (expressed in ml/min/mmHg) of carbon dioxide through the respiratory membrane in normal resting conditions is:

- a. 1200
- ☒ b. 400

- c. 65
- d. 21

33. The V/Q ratio in some areas of the lungs of one person equals zero because there is no ventilation and the blood flow remain normal. Which of the followings will be the values of alveolar partial pressures (in mmHg) of the respiratory gases in this situation?

- ☒ a. pO₂ 40 and pCO₂ 45
- b. pO₂ 104 and pCO₂ 45

- c. pO₂ 104 and pCO₂ 40
- d. pO₂ 40 and pCO₂ 0.3

34. Which of the following statements IS NOT correct regarding blood flow through the lungs and its distribution?

- a. The pulmonary blood flow is essentially equal to the cardiac output
- b. If the alveolar partial pressure of oxygen decreases the adjacent blood vessels constrict
- ☒ c. The blood flow is equally distributed to all areas of the lungs
- d. Some areas received more blood in the systolic period of the cardiac cycle

35. The decrease in oxygen affinity of Hb when the pH of blood falls, which facilitates its delivery to the tissues is:

- ☒ a. Bohr effect
- b. Haldane effect

- c. Haldane effect
- d. Haldane effect

36. Haldane Effect explains the influence of _____ on carbon dioxide transport.

- ☒ a. Oxygen
- b. Hemoglobin

- c. Bicarbonate
- d. Chloride ions

37. Which of the following areas of the respiratory centre controls voluntary forced exhalation and acts to increase the force of inspiration?

- ☒ a. Dorsal respiratory group
- b. Ventral respiratory group

- c. Neumotaxic centre
- d. Chemosensitive area

38. Central chemoreceptors are sensitive to: *increased arterial CO₂*
Decreased arterial pH

- a. pH of the extracellular fluid
- ☒ b. pH of its environment

- c. oxygen levels in the blood
- d. carbon dioxide levels in the blood

39. All of the followings lead to hyperventilation EXCEPT:

- ☒ a. increased pH in the blood
- b. decreased oxygen in the blood

- c. decreased pH in the cerebrospinal fluid
- d. increased carbon dioxide levels

40. Which of the followings IS NOT correct regarding the effect of oxygen on ventilation?

- a. Oxygen has low effect in its normal range

- b. It is only important in pronounced hypoxemia

- High Pa-O₂ strongly depress ventilation
41. The saturation percentage of the Haemoglobin in the blood leaving the lungs is around:
- 75
 - 97
 - 100
 - 104
42. Acute mountain sickness (altitude sickness) is caused by:
- Acute exposure to low levels of carbon dioxide
 - Acute exposure to low partial pressure of oxygen
 - Very prolonged exposure to high altitude
 - Acute exposure to high partial pressures of Nitrogen
43. Regarding the physiological significance of carbon dioxide and its role in the chemical regulation of the respiratory function, one of the following statements IS NOT correct.
- Maintains Pa-CO₂ within the normal range
 - Helps to maintain brain pH constant
 - Hypercapnia increases sensitivity to hypoxemia
 - Remains when other chemostimulation is lost
44. The exact function of one of the following areas of the respiratory center is still under discussion.
- Neumotaxic centre
 - Chemosensitive area
 - Dorsal respiratory area
 - Apneustic center
45. Carbon Dioxide is chiefly transported in blood:
- as dissolved CO₂
 - bound to hemoglobin
 - as bicarbonate ✓
 - in combination with plasma proteins
46. During physical exercise:
- Ventilation is usually greater than perfusion
 - Ventilation is usually lower than perfusion
 - Ventilation may be either greater or lower than perfusion
 - Ventilation and perfusion increase both in same proportion
47. One of the followings IS NOT considered to be a physiological factor which facilitates the diffusion of oxygen from the maternal to the fetal blood through the placenta.
- Higher affinity of fetal hemoglobin to oxygen compared with that of the mother
 - Greater gradient for diffusion of oxygen than that observed in the lungs of the mother
 - Greater concentration of Fetal hemoglobin to carry 50% more than the maternal Hb does
 - Double Bohr effect that makes easier the delivery of O₂ by maternal Hb and the uptake by fetal Hb
48. The respiratory distress syndrome of the newborn is caused by:
- Increased airways resistance
 - Reduced collapse pressure in both lungs
 - Increased release of pulmonary surfactant
 - Increased surface tension and reduced alveolar radius
49. Which of the followings is the cause of the deep-water blackout?
- rapid drop in the partial pressure of oxygen in the lungs on ascent
 - rapid drop in the partial pressure of carbon dioxide in the lungs on ascent
 - increase in the partial pressure of nitrogen in the lungs on ascent
 - decrease in the partial pressure of nitrogen in the lungs on ascent
50. The mean gradient for oxygen diffusion through the placental membrane to make possible the oxygenation of the fetal blood is about:
- 60
 - 50
 - 30
 - 20

ENDOCRINOLOGY

51. Which of the following is not a glycoprotein?
- Luteinizing hormone
 - ACTH
 - Follicle stimulating hormone
 - Thyroid stimulating hormone
 - Human chorionic gonadotropin
52. Which of the following is not true about Thyroid stimulating hormone?
- It stimulates thyroid to make and release thyroxine and triiodothyronine
 - Its release is stimulated by TRH from hypothalamus

It is inhibited by thyroid hormones through negative feedback
It is a steroid
It is formed by thyrotrope in the anterior pituitary gland

53. Which of the following is not true about adrenocorticotrophic hormone?

- a. It is a polypeptide with 39 amino acids
- b. Its target is the adrenal cortex
- c. It stimulates the release of corticosteroids

- ☒ d. Its release is stimulated by CRH from the pituitary gland
- e. It is inhibited by cortisol through negative feedback mechanism

54. Which of the following is not true about growth hormone?

- a. It is released in burst
- b. It has 191 amino acids
- c. It has half-life of 20 minute

- ☒ d. Its secretion is stimulated and inhibited by somatostatin
- ☒ e. It is secreted from the somatotropes of the adrenal medulla

55. Somatostatin inhibits the following except

- a. Insulin
- b. Glucagon
- c. Renin

- d. Gastrin
- ☒ e. None of the above

56. Growth hormone can be stimulated by the following except

- a. Hypoglycemia
- ☒ b. β -adrenergic agonist
- c. Dopamine

- d. Serotonin
- e. Thyroid hormones

57. Growth hormones can be inhibited by the following except

- a. Progesterone
- ☒ b. α -adrenergic agonist
- c. free fatty acids

- d. somatostatin
- e. supraphysiological concentration of cortisol

58. Which of the following is not true about the effect of growth hormone?

- a. It elicits growth of body cells
- b. It promotes protein deposition in tissues
- c. It enhances fat utilization for energy

- ☒ d. It increases carbohydrate utilization
- e. It stimulates cartilage and bone growth

59. Stress can be signaled by the following except

- a. Substrate lack
- b. Pain
- c. Trauma

- d. Hypovolemia
- ☒ e. Satiety

60. In response to stress, the following are released except:

- a. Catecholamine
- b. Angiotensin
- c. Growth hormone

- d. Glucagon
- ☒ e. None of the above

61. Which of the following is not true about adrenal glands?

- a. They are pyramid-shaped organs atop the kidneys
- b. They weigh 6-10 g
- ☒ c. Adrenal cortex is derived from the neural ectoderm
- d. Adrenal medulla is a modified sympathetic ganglion
- e. There are 2 glands in 1 organ

62. Which of the following is not a glucocorticoid?

- a. Corticosterone
- b. Prednisone
- c. Dexamethasone

- d. Hydrocortisone
- ☒ e. Aldosterone

63. The following are correct about the biological effect of cortisol except

- a. It stimulates gluconeogenesis during fasting
- b. It increases gluconeogenic enzyme and substrate in the liver
- c. It has catabolic effect on extrahepatic organs
- ☒ d. It decreases glycogen synthesis and accumulation in the liver
- e. It increases protein breakdown and decreases protein synthesis to provide amino acids for use by the liver

64. Which of the following is not true about the anti-inflammatory mechanism of cortisol?

- a. It inhibits migration of polymorphonuclear leukocytes, monocyte-macrophage and lymphocytes at inflammatory sites

- It inhibits the release of vasoactive proteolytic and other peptides
- It inhibits fibroblast growth and wound healing
- It decreases synthesis of macrocortin, which inhibits PLA₂, thus
- It decreases arachidonic acid

Which of the following is not correct about epinephrine?

- a. It increases heart rate
- b. It increases peristalsis
- c. It causes vasoconstriction in the skin and viscera
- d. It dilates the bronchioles
- e. It increases rate of cellular respiration

Which of the following is not a function of parathyroid hormone?

- a. It increases osteoclastic activity
- b. It increases renal tubular calcium resorption
- c. It decreases renal tubular resorption of phosphate
- d. It decreases 1-alpha hydroxylase activity
- e. It increases vitamin D level

Which of the following is not true about parathyroid gland?

- a. Its dimension is 3 x 6 x 2 mm
- b. It weighs about 35 mg
- c. It has chief cells
- d. There are 6 in human
- e. It is essential for life

High level of thyroid hormones can cause the following except

- a. Weight loss
- b. Palpitation
- c. Bradycardia
- d. Heat intolerance
- e. Muscle weakness

Which of the following is not true about thyroid gland?

- a. It consists of 2 lateral lobes joined by a midline isthmus
- b. The weight of the adult gland is about 10 – 25 g
- c. It synthesizes, stores and secretes thyroid hormones
- d. Its functional unit is the follicle
- e. The follicle is lined with columnar epithelial cells that contain thyroglobulin

Which of the following is not an inhibitor of thyroid stimulating hormone secretion?

- a. Somatostatin
- b. Cholecystokinin
- c. Glucocorticoid
- d. Androgens
- e. Estrogens

Mineralocorticoids stimulate sodium and potassium transport in the following except

- a. Sweat gland
- b. Salivary gland
- c. Intestinal epithelial cell
- d. Renal tubular cells
- e. None of the above

Hyperglycemia causes the following except

- a. Cellular hydration
- b. Glucosuria
- c. Osmotic diuresis
- d. Damage to the kidney
- e. Damage to the eyes

Glucagon performs the following functions except

- a. It inhibits fat storage
- b. It enhances strength of heart muscle
- c. It increases blood flow in the kidney
- d. It enhances bile secretion
- e. It increases gastric acid secretion

Which of the following is false

- Set point refers to a value range that the body works to maintain
- Processes that reduce discrepancies from the set point are known as negative feedback
- Homeostasis refers to the regulation of biological processes that keep certain variables within a fixed range
- Body temperature can be regulated by increasing internal energy use or by heat loss
- None of the above

Which of the following is false

- Basal metabolism is the energy used to maintain a constant body temperature while at rest
- Basal metabolic rate is the amount of energy expressed in calories that a person needs to keep the body functioning at rest
- Body temperature of poikilothermic species matches that of the environment
- Poikilothermic species lack the internal physiological mechanisms of temperature regulation
- Poikilothermic species are also known as warm-blooded animal

REPRODUCTION

76. During sex determination in fetal development in the absence of testis determining factor chromosome, the undifferentiated gonad develops into a:
- a) Testis
 - b) Fallopian tube
 - c) Upper part of the vagina
 - d) Ovary
 - e) Vulva
77. Determination of sex of a baby occurs at the moment of:
- a) Pregnancy
 - b) Lactation
 - c) Sexual intercourse
 - d) Conception
 - e) Ovulation
78. The genetic information responsible for the determination of a sex a baby is found at:
- a) The normal arm of the Y chromosome
 - b) The medium-short of the Y chromosome
 - c) The short arm of the X chromosome
 - d) The short arm of the Y chromosome
 - e) The long-short arm of the Y chromosome
79. Autosomes are:
- a) Sex determining chromosomes
 - b) Non-sex determining genes
 - c) Non-sex determining chromosomes
 - d) Non-sex determining elements in the testis
 - e) Non-sex determining cells in the testicular tissue
80. Testis determining factor is:
- a) An enzyme
 - b) A protein
 - c) Steroid
 - d) Glycoprotein
 - e) Pseudo-protein
81. Which of the following statement is CORRECT about sexual differentiation during fetal growth and development:
- a) The primitive gonads do not develop from the genital ridge in the abdomen during the embryological stage
 - b) In the absence of testes determining factor, this makes the cortex to atrophy
 - c) Testes determining factor is a hormone that determines whether the indifferent gonad develops into a testis
 - d) The Wolffian duct differentiates into the structures of the female genital tract
 - e) Mullerian inhibitory substance is secreted by Sertoli cells of fetal testes
82. Which of the following is NOT a component of urogenital sinus?
- a) Urogenital folds
 - b) Urogenital slit
 - c) Urogenital tubercles
 - d) Urogenital groove
 - e) Urogenital plate
83. The ACTIVE form of testosterone is:
- a) Bi-hydroxytestosterone
 - b) Hydroxytestosterone
 - c) Di-hydroxytestosterone
 - d) Di-troxytestosterone
 - e) Di-monotestosterone
84. The following are WRONG about psychological sex EXCEPT:
- a) Is a sex that is intended for procreation purpose
 - b) Is sex not intended for sensual pleasure and enjoyment
 - c) Individual indulge in psychological sex as a way of lessening their alienation, isolation and loneliness
 - d) There are emotional feelings attached in this kind sex
 - e) In one night, stand, when a male and female meeting for the first and indulge in sexual act is not an example of psychological sex
85. Control of onset of puberty is initiated by:
- a) Release of hypothalamic corticotropin releasing hormone
 - b) Release of hypothalamic thyrotropin releasing hormone
 - c) Release of hypothalamic gonadotrophic releasing hormone
 - d) Release of hypothalamic growth hormone releasing hormone
 - e) Release of prolactin releasing hormone
86. The following are TRUE about ambiguous genitalia EXCEPT:
- a) Is not a disorder of sexual differentiation
 - b) It is a condition where the external genitalia do not have the typical anatomical appearance of normal male or female genitalia
 - c) It can be classified into true hermaphroditism and pseudo-hermaphroditism
 - d) A true hermaphrodite possesses both ovarian and testicular tissues, either separately or combined form known as ovotestis
 - e) A pseudo-hermaphrodite is an individual with the genetic composition and gonads of one sex and the genitalia of other

87. Which of the following is NOT a secondary or accessory organ of reproduction?
 a) The epididymis
 b) The vas deferens
 c) The ejaculatory duct
 d) ☒ Testis
 e) The penis
88. The following are external genitalia in the female EXCEPT:
 a) The mons pubis
 b) The pons pubis
 c) The labia majora
 d) The labia minora
 e) The clitoris
89. The following constitutes the vulva EXCEPT:
 a) The labia majora and
 b) The labia minora
 c) The bulb and glans of
 d) The clitoris
 e) ☒ The vagina
90. Which of the following secretes the fibrinolysin into semen?
 a) Seminal vesicle
 b) Epididymis
 c) Cowper's gland
 d) ☒ Prostate gland
 e) Ejaculatory gland
91. The penis is made up of:
 a) Four long cylindrical masses of erectile tissue
 b) Two long cylindrical masses of erectile tissue
 c) ☒ Three long cylindrical masses of erectile tissue
 d) Five long cylindrical masses of erectile tissue
 e) One pair of long cylindrical masses of erectile tissue
92. Sperms are stored and continue their maturation in the:
 a) Vas deferens
 b) Ejaculatory duct
 c) ☒ Epididymis
 d) Seminiferous tubules
 e) Seminal vesicle
93. Which of the following produces a whitish mucus that accompanies semen?
 a) ☒ Seminal vesicle
 b) Ejaculatory duct
 c) Epididymis
 d) Seminiferous tubules
 e) Cowper's gland
94. Which of the following is the primary organ of reproduction in the female?
 a) ☒ The ovaries
 b) The fallopian tubes
 c) The uterus
 d) Vagina
 e) The vulva.
95. The labia majora and clitoris in a female are homologous to the:
 a) Scrotum and testis respectively in a male
 b) Scrotum and urethra respectively in a male
 c) ☒ Scrotum and penis respectively, in a male
 d) Scrotum and ejaculatory duct in a male
 e) Scrotum and prostate in a male
96. The following are ovarian hormones EXCEPT:
 a) Estrogen
 b) Progesterone
 c) Inhibin
 d) Relaxin
 e) ☒ Luteinizing hormone
97. The following are TRUE about the uterus EXCEPT:
 a) It is formed of a body and a cervix
 b) The body wall is formed of three layers
 c) The myometrial layer of the uterus contains cells that are capable of increasing in size
 d) Non-pregnant state = 7cm long, 4 cm wide and 2.5 cm thick and between 30-50g
 e) ☒ It expands during pregnancy to about 2000g to 2200g
98. The following are features of puberty in the male EXCEPT:
 a) Enlargement of testes
 b) Hyperpigmentation of the scrotum plus rugae folds
 c) Increased in the length and thickness of the penis
 d) Growth of accessory reproductive organs
 e) ☒ Last ejaculation
99. The following are features of puberty in the female EXCEPT:
 a) Appearance of hair in the pubic region in a characteristic flat-topped pattern
 b) Deposition of adipose tissue in the buttocks and thigh
 c) Pigmentation of the areolae and nipples increases somewhat
 d) ☒ Low pitched and thin voice
 e) Females develop less body hair than males, but more scalp hair is present
100. Concerning spermatogenesis, the following are true EXCEPT:
 a) Spermatozoa are formed from the primitive germ cells in the walls of loops of convoluted seminiferous tubules
 b) ☒ Begins at puberty and continues up till 100 years and stops ✓
 c) Occurs as result of stimulation by gonadotropic hormones from the anterior pituitary
 d) The period of spermatogenesis, from spermatogonia to spermatozoa, takes about 74 days
 e) The processes of spermatogenesis have basically three stages

SHORT ANSWER QUESTIONS. ANSWER ALL QUESTIONS IN THE SPACES PROVIDED ONLY. (30 MARKS TOTAL)
Write a short note on the following: (15 MARKS)

Physiology of lactation

(b) Puberty

(c) Hormonal control of testicular function

List and explain 15 physiological effects of cortisol (15 marks)

- Stimulates gluconeogenesis during fasting
- It has anabolic effect on the liver & induces gluconeogenic enzyme induce gluconeogenic substances
- Has catabolic effect
- Increases glycogen synthesis & accumulation in the liver
- Increases protein breakdown & ↓ protein synthesis to provide aa for use by the liver
- ↓ glucose and aa uptake in the periphery
- ↑ lipolysis and enhances the effect of lipolytic stimulation such as catecholamines
- Increases capillary fragility
- Inhibits linear growth and skeletal maturation in children
- Required for breast development
- Required for normal growth
- ↑ glutamine synthesis in neural retina
- Increases hepatic enzyme in the fetal liver

SECTION C: LONG ANSWER QUESTIONS. ANSWER ALL QUESTIONS IN THE SPACES PROVIDED ONLY. (20 MARKS TOTAL)

1. List 10 peptide and 10 steroid hormones (10 marks)

PEPTIDE HORMONES (0.5 Marks each)	STEROID HORMONES (0.5 Marks each)
TRH (Thyrotropin releasing hormone) - ↑ TSH & Prolactin	Cortisol (gluco-corticoid)
CRH (Corticotropin releasing hormone) - ↑ ACTH	Aldosterone
Somatostatin - ↓ secretion of GH	ADT & Rostenedione (Androgens)
GHRH (Growth hormone releasing hormone) - ↑ secretion of GH	Testosterone
TSH (Thyroid stimulating hormone)	Estrogen
FSH (Follicle stimulating hormone)	Progesterone
LH (Luteinizing hormone)	Progesterone
GH (Growth hormone)	Estriol
Prolactin	1,25-dihydroxy-cholecalciferol
ACTH - Adrenal cortical hormone	
MSH - (Melanin synthesizing hormone)	

② Biceps Mechanism of Erection

Short hormone releasing hormone